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#include <stdio.h>
#include <stdlib.h>
#define MAX 64
 struct Stack {
int top;
int items[MAX];
};
void initStack(struct Stack* s) {
s->top = -1;
}
int isFull(struct Stack* s) {
return s->top == MAX - 1;
}
int isEmpty(struct Stack* s) {
return s->top == -1;
}
void push(struct Stack* s, int value) {
if (isFull(s)) {
printf("Stack Overflow!\n");
return;
```

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}
s->items[++(s->top)] = value;
}
int pop(struct Stack* s) {
if (isEmpty(s)) {
printf("Stack Underflow!\n");
return -1;
}
return s->items[(s->top)--];
}
void decimalToBinary(int num) {
struct Stack s;
initStack(&s);
if (num == 0) {
printf("Binary equivalent: 0\n");
return;
}
int originalNum = num;
while (num > 0) {
push(&s, num % 2);
num = num / 2;
}
```

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printf("Binary equivalent of %d: ", originalNum);
while (!isEmpty(&s)) {
printf("%d", pop(&s));
}
printf("\n");
}
int main() {
int decimalNumber;
printf("Enter a decimal number: ");
if (scanf("%d", &decimalNumber) != 1) {
printf("Invalid input.\n");
return 1;
}
if (decimalNumber < 0) {
printf("Please enter a non-negative integer.\n");
return 1;
}
decimalToBinary(decimalNumber);
return 0;
}
```